[INCH-POUND]
A-A-58089
10 DECEMBER 1996

COMMERCIAL ITEM DESCRIPTION

COMPARATOR, BENCH TYPE, ELECTRONICALLY AMPLIFIED

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-C-83988 and MIL-C-83997 for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers electronically amplified bench type comparators.

1.2 Intended use. Comparators covered by this CID will be used for determining ultra-precision dimensional values and accuracy's of master inspection rings, disc, gage blocks, and for determining accuracy's of other measuring devices.

2. CLASSIFICATION

2.1 Comparators covered by this CID shall be of the following types, styles, classes and sizes. The types, styles, class and sizes shall be as specified (see 7.3).

Type I - Ring and Discs

Style 1 - Horizontal Measurement

Class A - Computer Interface
Class B - No Interface

Size 1 - .040″ to 12.000″ I.D.
0″ to 10.750″ O.D.

Size 2 - .040″ to 13.250″ I.D.
0″ to 12.250″ O.D.

Type II - Gage Block

Style 2 - Vertical Measurement

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: WR-ALC/LKJE, 460 Second Street, STE 221 Robins AFB GA 31098-1640

AMSC N/A FSC 5220

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Class A - Computer Interface
Class B - No Interface

Size 3 - 0″ to 4.00″ Range
Size 4 - 0″ to 24.00″ Range

3. SALIENT CHARACTERISTICS

3.1 General requirements. The comparator shall be new and of the quality and design currently being offered by manufactures as a industry excepted standard. The comparator shall be a self contained portable unit with digital readout which shall be an integral part of the assembly. The comparator shall be capable of a full range of repetitive gaging accuracy's comparison to within .000001″. A floating frame design or system equal to or better shall be utilized to provide a true high sensitive point to point measurement with the reliable simplicity of a single electronic gage head. The comparator may provide computer interface in accordance with EIA RS-232.

3.2 Construction. Construction shall be free from any defects which would adversely affect the performance or maintainability of the individual components or overall assembly. The comparator shall be complete so that when assembled and connected to the specified source of power it shall be ready for operation.

3.3 Electrical requirements. The comparator shall be a self contained portable unit designed to operate from a dual 115/220 volts AC, 50/60 Hz. Unless otherwise specified (see 7.3), the comparator shall be initially wired for 115 volts operation. The power supply cord and its plug shall comply with requirements of UL 45.

3.4 Electronic amplifier. The amplifier shell be solid state and meet all current electronic gaging industry standards. It shall have two or more electronic stepper-switches to meet the requirements specified here in.

3.5 Range and scale divisions. The amplifiers shall meet or exceed the range and scale divisions specified in Table I.

TABLE I . Range and scale divisions.

Amplifier Scales	Measurement	
	Range	Per Division
High	± 0.00002″	0.000,001″
Low	± 0.0002″	0.000,010″

3.6 Coatings or painting. Corrosion resistant treatment, painting, or plating shall be in accordance with manufacturer's standard commercial practice with the exception that there shall be no cadmium plating used on any component of the assembly.

3.7 Components and accessories. The comparator components and accessories shall be as specified (see 7.3).

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3.8 Nameplate. A corrosion resistant nameplate, permanently and legibly marked with the following information shall be securely attached to a component of the comparator assembly.

Nomenclature
Manufacturer's Model Number
Manufacturer's Serial Number
Stock Number
Manufacturer's Name
Contract Number
Special Characteristics (volts, watts, etc.)
Date of Manufacture
U. S.

4. REGULATORY REQUIREMENTS

4.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirement, and promotes economically advantageous life cycle costs.

4.2 Environmental protection. The comparator shall meet all applicable Environmental Protection Agency restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health and shall be met during the manufacturing, service, transportation, storage, and operation of the comparator.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product Conformance. The Comparator provided shall meet the salient characteristics of this commercial item description, conform to the manufacture's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

6. PACKAGING

6.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be as specified in the contract or purchase order (See 7.3).

7. NOTES

7.1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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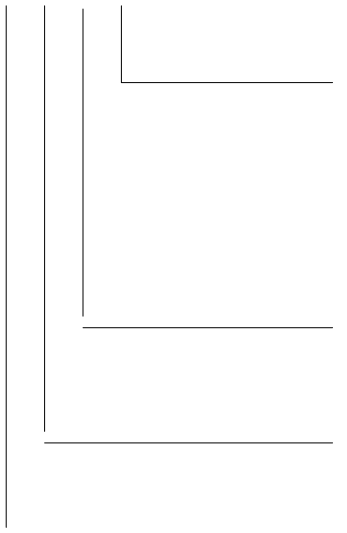
This example describes a part numbering system for CID A-A-58089.

AA58089 - 1 A 2 B

Example of reference part number:
AA58089 - 1A2B



Size: A - Size 1 - .040″ to 12.000″ I.D.
0″ to 10.750″ O.D.
B - Size 2 - .040″ to 13.250″ I.D.
0″ to 12.250″ O.D.
C - Size 3 - 0″ to 4.00″ Range
D - Size 4 - 0″ to 24.00″ Range

Class: 1 - Class A - Computer Interface
2 - Class B - No Interface

Style: A - Style 1 - Horizontal Measurement
B - Style 2 - Vertical Measurement

Type: 1 - Type I - Ring and Discs
2 - Type II - Gage Block

7.2 Addresses for obtaining copies of referenced documents.

UNDERWRITERS LABORATORIES (STANDARDS FOR SAFETY)

UL 45 - Portable Electric Tools

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006-5009.)

7.3 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document
b. Type, style, class and size comparators as required (see 2.1)
c. Electrical power requirements, if different (see 3.3)
d. Components and accessories as specified (see 3.7)
e. Packaging requirements, if required (see 6.1)

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7.4 National Stock Numbers (NSNs). The following is the only NSN assigned which correspond to this CID. There are currently 82 NSNs with the same Approved Item Name (AIN) that may be indicative of all possible NSNs, associated with the CID.

5220-00-973-6333

7.5 Known sources. The following is a list of sources known to meet the requirements of this CID. The list may not be indicative of all possible sources.

STARRETT TOOLS
STANDARD GAGES
FEDERAL PRODUCTS

7.6 Subject term (key word) listing:

Comparator
Digital readout
Floating frame design
Gage blocks
Gage head, single electronic
Point to point measurement
Rings and Disc
Stepper-switch , electronic

MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AR
Navy - AS

Rrviewers:
Army - AT, AV, ME, MI

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
Air Force - 84

Agent Activity:
Air Force - 99

(Project 5220-0229)

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